USN. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22ESEC22]

SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2023

Data Structures and Algorithms using C

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

PART-A **05 X 02=10**

1.	a.	Write the structure definition of single linked list in C language.	L1	CO1	02
	b.	What are the two basic operations associated with stacks.	L1	CO2	02
	c.	Write any two applications of stack.	L1	CO2	02
	d.	What is tree and write a structure of tree.	L1	CO4	02
	e.	Write a representation of strongly connected graph.	L1	CO4	02
		PART-B **06 x 05 = 30**			
2.	a.	Write a C function of singly linked list with 'n' number of nodes.	L3	CO1	05
		OR			
	b.	Write a short note on ADT.	L2	CO1	05
3.	a.	Outline the differences between array and linked list.	L3	CO1	05
		OR			
	b.	Write a C function of insert Queue() & delete Queue() function using array method.	L3	CO3	05
4.	a.	Discuss the various operations performed on the Queue and illustrate with an example.	L2	CO3	05
		OR			
	b.	Write a C program for insert Queue function using linked list method.	L3	CO3	05
5.	a.	Explain Kruskal's algorithm and the steps to finding the MST.	L2	CO4	05
		OR			
	b.	Outline the different properties of binary trees.	L2	CO4	05
6.	a.	Explain briefly different tree terminologies.	L2	CO4	05
		OR			
	b.	List out any five application of binary tree.	L1	CO5	05
7.	a.	Write a C program to implement linear search algorithm.	L3	CO5	05
		OR			

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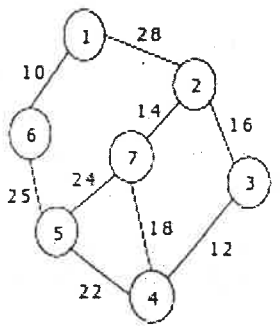
	b.	Illustrate the functions that are necessary to implement a student record management system.	L2	CO6	**05**
		PART-C **06 X 10=60**			
8.	**a.**	Develop a C function and write the steps for double linked List with creating n number of nodes.	L3	CO1	**10**
		OR			
	b.	What is a data structure? Explain primitive & non-primitive data structures with an example.	L2	CO1	**10**
9.	**a.**	Explain contiguous structures & non-contiguous structures with an example.	L2	CO1	**10**
		OR			
	b.	Define stack. Explain the PUSH & POP operation with an example.	L2	CO2	**10**
10.	**a.**	Discuss the two problems associated in linear queue and explain the representation of circular queue with an example.	L2	CO3	**10**
		OR			
	b.	Analyze the Procedure to convert from infix to postfix expression for the expression $((A-(B+C))*D)\uparrow(E+F)$.	L4	CO2	**10**
11.	**a.**	Write a C program to perform the following operations in binary tree a) Create a complete binary tree using array method b) Inorder traversal	L3	CO4	**10**
		OR			
	b.	Explain the different properties and write a representation of full binary tree and complete binary tree.	L2	CO4	**10**
12.	**a.**	Construct the minimal spanning tree for the graph shown below.  Figure 12(a)	L3	CO4	**10**
		OR			

	b.	Write a C program and illustrate bubble sort algorithm with an example.	L3	CO5	**10**
13.	**a.**	Write a C program and illustrate linear search with an example.	L3	CO5	**10**
		OR			
	b.	With a flowchart/algorithm write a C program to insert a student record in a student database management system.	L3	CO6	**10**